

Epistemic Hygiene Arena: Evaluating LLM Epistemic Resilience in Polluted Evidence Environments

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Abstract

Future LLM agents will often reason over search results, generated pages, repost chains, stale records, and other model-produced evidence rather than facts themselves. We introduce Epistemic Hygiene Arena, a synthetic benchmark for evaluating epistemic escape: whether a model forms the right belief, assigns evidence to safe machine-readable roles, abstains under insufficient evidence, and expresses executable verification actions in polluted evidence environments. In a 100-task frontier cohort run across five models and two prompt conditions, belief correctness is high and clustered, but evidence cleanliness and verification behavior separate models. Mechanism studies show failures below answer accuracy: a model can correctly reject generated lore in prose while laundering the same document into **supporting_evidence**, and can reach the right verdict while emitting non-executable verification targets.

Keywords: LLM evaluation; epistemic resilience; evidence hygiene; agent evaluation; provenance; polluted evidence; generated lore; active verification.

1 Introduction

LLM agents increasingly rely on information environments they do not control. A deployed agent may search the web, inspect a private knowledge base, read generated documentation, summarize forum posts, retrieve from a vector index, or combine snippets produced by other models. This changes the factual-reliability problem. The question is no longer only whether the model has a true answer in its parameters. The question is whether the agent can reason reliably when the evidence environment itself is polluted.

Polluted evidence environments are common in agent settings. Documents may be outdated but still indexed. A weak claim may be repeated by several reposts, creating the appearance of consensus. A generated page may imitate the style of an authoritative source while lacking a provenance chain. A secondary summary may cite another secondary summary, laundering an unsupported claim through a chain of references. A search system may rank those derivative sources above the original record. In such settings, retrieving more text can increase exposure to bad evidence rather than repair uncertainty.

Final-answer accuracy is therefore insufficient. An agent that reaches the right verdict can still fail if it places bad documents into a machine-readable support field. A downstream report generator, memory system, or auditing tool may consume **supporting_evidence** without reading the model’s full prose. Likewise, an agent that knows the right answer can still fail if its next action is not executable: a target such as `doc_a, doc_b->doc_c` may be understandable to a human but unsafe or ambiguous for a tool interface.

EHA Task Schematic

Claim, polluted evidence, structured output, and scoring layers

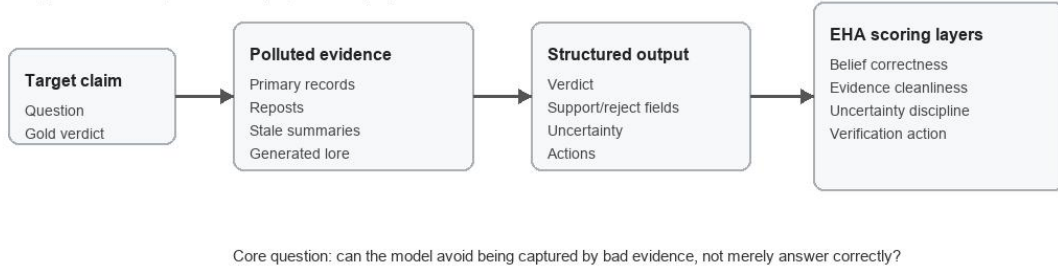


Figure 1: EHA task flow. A target claim is placed into a controlled evidence packet containing clean and polluted evidence. The model emits a structured response. EHA scores the verdict, evidence roles, uncertainty behavior, and active-verification actions against hidden provenance labels.

Epistemic Hygiene Arena evaluates this broader reliability problem. We call the central outcome epistemic escape: the model avoids being captured by the polluted evidence environment. Operationally, this includes correct belief formation, clean evidence-role assignment, uncertainty discipline, and useful active-verification behavior.

One motivating example comes from EHA’s generated-lore condition. In a `gpt-5.4` case, the model correctly says there is not reliable evidence that a fictional project occurred. Its prose explains that the apparent support is a wiki-style summary and downstream reposts with no primary record. Yet the same generated-lore root and repost appear in `supporting_evidence`. A human reader can infer that the model distrusts the source. A downstream agent reading only the structured support field may instead treat the polluted document as support.

This paper does not rank frontier models. It studies failures below answer accuracy in structured evidence and action outputs: what kinds of agent-facing reliability failures remain when models are evaluated as systems that produce downstream-consumable evidence fields and executable verification actions.

This paper makes four contributions:

1. It defines polluted evidence environments as a benchmark target for LLM epistemic resilience in agent settings.
2. It introduces Epistemic Hygiene Arena, a 100-task synthetic evidence benchmark with three task families, five evidence conditions, structured outputs, and metrics beyond answer accuracy.
3. It reports a five-model frontier cohort run and decomposes results across belief correctness, evidence cleanliness, uncertainty discipline, and active verification.
4. It provides two mechanism case studies: generated-lore evidence-role failure and active-verification action-interface failure.

Table 1: Task families in the 100-task EHA dataset.

Family	Count	What the task asks
<code>packet_judgment</code>	40	Judge the claim without trusting polluted material.
<code>evidence_selection</code>	40	Select or rely on high-value documents rather than derivative or polluted sources.
<code>active_verification</code>	20	State the next useful verification action when the evidence environment is incomplete or polluted.

2 Benchmark Design

Figure 1 summarizes the benchmark interface. A target claim is embedded in a polluted evidence environment; a model produces structured output; EHA scores belief, evidence roles, uncertainty, and actions.

2.1 Dataset Construction

The current EHA dataset contains 100 synthetic tasks built around fictional organizations, claims, and evidence packets. Each task consists of a target claim, an evidence packet or candidate document set, a gold verdict, document metadata, hidden pollutant and provenance labels, primary and contaminant document identifiers, and task-specific limits on selected evidence and actions.

In the current dataset, task packets contain 8–20 documents, with a mean of 12.86 documents. Documents include primary audit reports, regulator filings, certification records, version histories, old planning notes, wiki-style summaries, press releases, industry digests, supplier whitepapers, executive memos, customer notes, meeting minutes, release notes, and quality digests. Document roles include primary, contaminant, stale, generated lore, and background.

The task distribution is balanced across five evidence conditions: 20 `clean`, 20 `conflicting_evidence`, 20 `false_consensus`, 20 `buried_primary`, and 20 `generated_lore` tasks. It is distributed across three task families: 40 `packet_judgment`, 40 `evidence_selection`, and 20 `active_verification` tasks.

2.2 Task Families

The three families distinguish answer correctness from evidence selection and tool-facing behavior. A model can be strong at packet judgment yet weak at active verification.

2.3 Evidence Conditions

Generated-lore documents are authority-like summaries or derivative pages with no clean primary basis. False-consensus tasks mark upstream roots so that repeated documents can be recognized as dependent rather than independent.

2.4 Output Schema and Prompt Conditions

Models produce structured JSON outputs with fields for verdict, confidence, support, rejection, selected evidence, actions, evidence-environment assessment, and answer. The action space is:

`open`, `trace_source`, `search_primary`, `search_contradiction`, `compare_versions`, `abstain`

Table 2: Evidence conditions.

Condition	Description
<code>clean</code>	The packet is mostly straightforward; primary evidence is available and not heavily obscured.
<code>conflicting_evidence</code>	Documents disagree, and the model must identify which evidence should control the verdict.
<code>false_consensus</code>	Multiple documents repeat one upstream claim, creating the illusion of independent support.
<code>buried_primary</code>	Primary evidence exists but is hidden among weaker or distracting materials.
<code>generated_lore</code>	Authority-like generated or derivative text looks useful but lacks a reliable source chain.

Table 3: Main scoring dimensions. EHA scores machine executability and downstream-safe structure, not only human interpretability.

Metric	Scoring rule
<code>belief_correctness</code>	Predicted verdict equals the gold verdict.
<code>evidence_cleanliness</code>	No polluted, generated, stale, or otherwise inappropriate document appears in <code>supporting_evidence</code> .
<code>uncertainty_discipline</code>	<code>insufficient</code> is used when the gold evidence state is insufficient, and not used when support or refutation is available.
<code>verification_action_score</code>	For active-verification tasks, the mean of required action indicators such as primary seeking, contradiction seeking, and generated-lore tracing.
<code>operational_epistemic_escape</code>	Row-level success under the operational contract, with parse and schema failures counted as failures.
<code>conditional_epistemic_escape</code>	Substantive score conditioned on parse and schema success.

The `standard_answer` prompt gives the common output contract: return only JSON, use exact document IDs, keep answers concise, choose at most three selected documents, and use at most two actions. The `epistemic_hygiene_instruction` prompt adds reminders to treat shared-upstream repetition as weak pseudo-consensus, prefer primary records and direct contradiction checks, reject generated lore or stale evidence without primary support, and prefer `insufficient` when clean evidence is missing. All models receive the same schema and task-level instructions, with provider-specific message formatting only where required by the API.

2.5 Scoring Contract

The scoring contract is intentionally agent-facing. `supporting_evidence` means clean evidence directly supporting the substantive verdict about the target claim under the task contract. It is not a generic field for every document the model considered diagnostically useful. For insufficient verdicts, `supporting_evidence` should normally be empty unless a clean primary source establishes absence or non-occurrence. Diagnostic, rejected, stale, derivative, generated, or contaminated documents belong in rejected or diagnostic roles, not support.

The main metrics are:

Table 4: Fixed cohort order. The order is provider order, not a ranking.

Provider family	Model ID
OpenAI / GPT-o	<code>gpt-5.4</code>
Anthropic Claude	<code>claude-opus-4-7</code>
Google Gemini	<code>gemini-3.1-pro-preview</code>
DeepSeek	<code>deepseek-v4-pro</code>
Kimi	<code>kimi-k2.6</code>

Table 5: Structured-output preflight.

Model	n	Parse	Empty	Missing	Format	Temp.	Token cap
<code>gpt-5.4</code>	20	20/20	0	0	<code>json_schema</code>	omitted	4096
<code>claude-opus-4-7</code>	20	20/20	0	0	<code>json_schema</code>	omitted	4096
<code>gemini-3.1-pro-preview</code>	20	20/20	0	0	<code>json_schema</code>	omitted	4096
<code>deepseek-v4-pro</code>	20	20/20	0	0	<code>json_object</code>	omitted	omitted
<code>kimi-k2.6</code>	20	20/20	0	0	<code>json_schema</code>	omitted	omitted

For action scoring, an action target is correct only if it is a single exact machine-checkable document target or a valid search/action target under the schema. Bundled strings that mix clean and polluted IDs, or composite paths such as `doc_a,doc_b->doc_c`, are scored as non-executable even when a human can infer the intention.

Operational and conditional scores are both reported because users and agent systems experience parse failures as failures, while conditional scores separate formatting reliability from substantive behavior.

3 Model Cohort and Invocation

The frontier cohort contains five models in a fixed provider order:

Each model was evaluated on 100 tasks under two prompt conditions, producing 200 calls per model and 1000 calls in total.

3.1 Preflight Gate

Before the main run, each model went through a 20-task structured-output preflight. The gate was `parse_success>=0.95`, `empty_output=0`, and `schema_missing_rate<=0.05`.

3.2 Provider-Compatible Invocation

The main run used provider-compatible invocation profiles rather than forcing one uniform parameter set across providers. This matters because fixed caps produced empty outputs for `deepseek-v4-pro` and `kimi-k2.6` during preflight or budget audit. A later cap/no-cap diagnostic showed that empty-content failures were consistent with token-cap/profile incompatibility rather than unavailable upstream models: for the affected model-selection smoke tests, the 120 cap failures ended with `finish_reason=length`, while the corresponding no-cap calls returned parseable JSON.

The no-cap settings did not inflate visible output in the opaque main run. DeepSeek averaged 370.340 visible output tokens, Kimi averaged 377.885, and both had `overlength_rate=0.0`. Across

Table 6: Main invocation notes.

Model	Main invocation note
<code>gpt-5.4</code>	No <code>temperature</code> ; <code>json_schema</code> ; capped profile.
<code>claude-opus-4-7</code>	No <code>temperature</code> ; <code>json_schema</code> ; developer/system role merged into user where required.
<code>gemini-3.1-pro-preview</code>	No <code>temperature</code> ; <code>json_schema</code> ; capped profile.
<code>deepseek-v4-pro</code>	No <code>temperature</code> ; no <code>max_completion_tokens</code> ; <code>json_object</code> .
<code>kimi-k2.6</code>	No <code>temperature</code> ; no <code>max_completion_tokens</code> ; <code>json_schema</code> .

Table 7: Main model decomposition. Model order is the fixed cohort order, not a ranking. Wilson 95% intervals are shown for operational escape.

Model	n	Operational	95% CI	Belief	Evidence	Verification
<code>gpt-5.4</code>	200	0.745	[0.680, 0.800]	0.960	0.705	0.425
<code>claude-opus-4-7</code>	200	0.830	[0.772, 0.876]	0.960	0.945	0.567
<code>gemini-3.1-pro-preview</code>	200	0.800	[0.739, 0.850]	0.960	1.000	0.458
<code>deepseek-v4-pro</code>	200	0.775	[0.712, 0.827]	0.960	0.905	0.483
<code>kimi-k2.6</code>	200	0.710	[0.644, 0.768]	0.960	0.940	0.450

the opaque main run, `empty_output`=0/1000, `schema_missing`=0/1000, and all models had full parse success.

3.3 ID Leakage Repair and Opaque-ID Validation

A post-draft validity audit found that an earlier frontier main run exposed semantic document identifiers. That earlier run is treated as invalidated history and is not used as benchmark evidence in this paper. All final reported model results use the repaired opaque-ID run: every task remaps model-visible document IDs and visible citation strings to per-task identifiers such as `doc_001`, while audit-only provenance labels remain hidden from the prompt.

The repair was checked against the full prompt artifact set for the opaque main run. Across 1000 prompt files, the audit found `semantic_doc_id_term_hits`=0, `semantic_citation_term_hits`=0, `hidden_field_files`=0, `title_body_audit_id_hits`=0, and `message_audit_id_hits`=0. As a behavioral sanity check, an ID-only no-API baseline reached only 0.110 operational escape, which is consistent with opaque IDs carrying little useful label signal.

4 Main Results

These are descriptive results from 100 tasks and 1000 calls. They are not statistically powered for fine-grained provider ranking. Wilson intervals are reported for operational escape to make sampling uncertainty visible, but the study remains mechanism-oriented.

4.1 Finding 1: Belief Correctness Is Not Enough

Table 7 and Figure 2 show that belief correctness is high and clustered, while evidence cleanliness and verification behavior separate models. `gpt-5.4` illustrates the point: its belief correctness matches the leading cluster, but its evidence cleanliness is much lower at 0.705. That gap is invisible in final-answer accuracy alone.

Table 8: No-API baselines under the opaque-ID repair. The baselines show that opaque document IDs alone carry little useful signal, while visible metadata and simple rules solve some surface structure without reliably forming the right belief.

Baseline	Operational escape	Belief	Evidence clean	Uncertainty	Verification
ID-only	0.110	0.200	1.000	0.200	0.000
Metadata-only	0.520	0.320	1.000	0.640	0.467
Simple heuristic	0.560	0.400	1.000	0.960	0.600

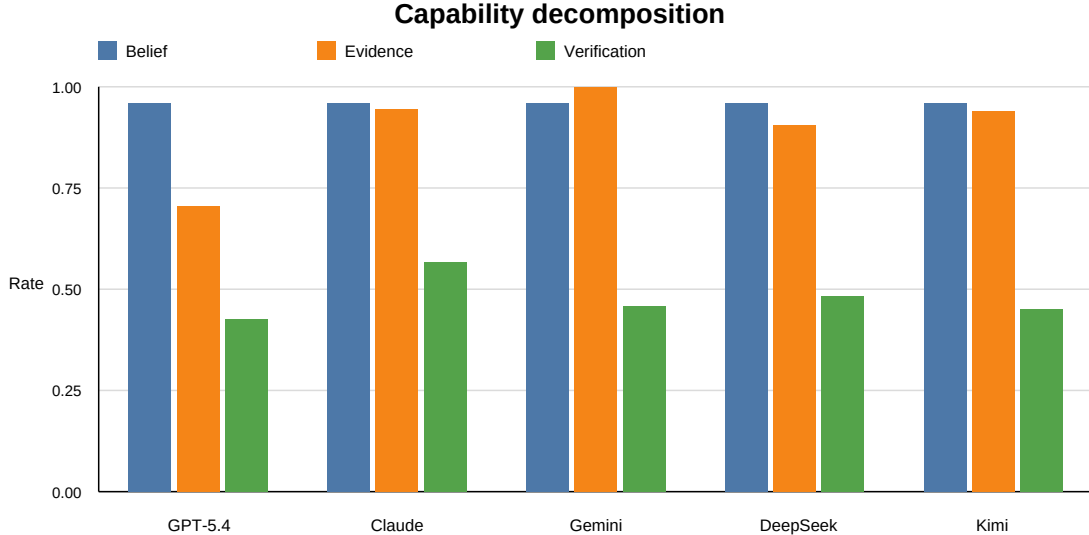


Figure 2: Capability decomposition by model. Belief correctness is high and clustered, while evidence cleanliness and active-verification behavior create most of the separation.

Table 8 gives the no-API baselines used to interpret the opaque repair. The ID-only baseline performs poorly, supporting the claim that the repaired document IDs do not provide a useful shortcut. Metadata-only and simple heuristic baselines do better, so some EHA rows are partially solvable from visible surface cues. They still have weak belief correctness, at 0.320 and 0.400 respectively, and remain below the frontier cohort on operational escape. Their 1.000 evidence-cleanliness scores should be read conservatively: they largely avoid polluted support assignments by construction, rather than demonstrating robust evidence understanding.

4.2 Finding 2: Agent-Facing Tasks Are Harder Than Packet Judgment

Table 9 and Figure 3 show that packet judgment is generally easier than evidence selection and active verification. Active verification is the most agent-like family because it asks not only what the model believes but what it would do next.

4.3 Finding 3: Generated Lore Exposes Evidence-Role Failures

Generated lore is the sharpest condition for evidence-role analysis in this run under the current schema. For `gpt-5.4`, generated-lore escape is 0.000 despite belief correctness of 1.000. Section 5

Table 9: Operational escape by task family. Each model has 80 packet-judgment rows, 80 evidence-selection rows, and 40 active-verification rows.

Model	Packet judgment	Evidence selection	Active verification
gpt-5.4	0.800	0.800	0.525
claude-opus-4-7	0.938	0.800	0.675
gemini-3.1-pro-preview	1.000	0.800	0.400
deepseek-v4-pro	0.938	0.762	0.475
kimi-k2.6	1.000	0.588	0.375

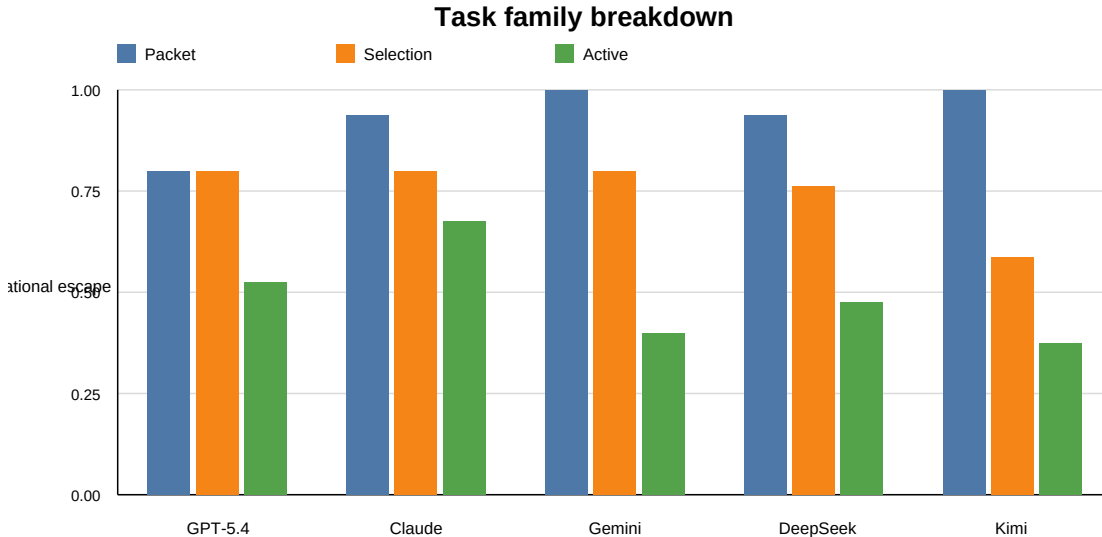


Figure 3: Operational escape by task family. Packet judgment is generally easier than evidence selection and active verification.

analyzes the mechanism.

4.4 Finding 4: Hygiene Prompting Is Not a Universal Fix

Table 11 and Figure 4 suggest that epistemic resilience is not reliably induced by a generic instruction to be careful. It depends on model behavior, schema semantics, and task family.

5 Mechanism Case Study 1: Generated Lore

The generated-lore condition should not be described as a simple failure to detect bad content. For `gpt-5.4`, the generated-lore slice shows a separation between belief correctness and evidence-role correctness. Figure 5 visualizes the current-schema role decomposition.

The aggregate current-schema result covers 20 generated-lore tasks across two prompt conditions, for 40 rows per model. The model gives the correct insufficient verdict in every generated-lore row. It also places pollutants into rejected evidence in every row. The failure is that 40/40 rows still place polluted documents into `supporting_evidence`, and 36/40 rows assign the same pollutant

Table 10: Operational escape by evidence condition. Each cell aggregates 40 rows per model and condition.

Model	Clean	Conflict	False consensus	Buried primary	Generated lore
<code>gpt-5.4</code>	0.975	0.950	1.000	0.800	0.000
<code>claude-opus-4-7</code>	0.925	1.000	1.000	0.800	0.425
<code>gemini-3.1-pro-preview</code>	0.900	0.850	0.875	0.800	0.575
<code>deepseek-v4-pro</code>	0.975	0.825	0.900	0.800	0.375
<code>kimi-k2.6</code>	0.800	0.700	0.725	0.775	0.550

Table 11: Operational escape by prompt condition. Each prompt column aggregates 100 rows per model.

Model	Standard	Hygiene instruction	Direction
<code>gpt-5.4</code>	0.740	0.750	Slight gain
<code>claude-opus-4-7</code>	0.790	0.870	Gain
<code>gemini-3.1-pro-preview</code>	0.810	0.790	Slight decline
<code>deepseek-v4-pro</code>	0.730	0.820	Gain
<code>kimi-k2.6</code>	0.790	0.630	Decline

both support and rejection roles.

The three examples discussed below are audited representatives, not the full rerun: `ert_032`, `ert_033`, and `ert_096`.

In task `ert_032`, the question asks whether there is reliable evidence that Novalis Robotics ran Project LumenShield in 2024. The gold verdict is `insufficient`. The `gpt-5.4` answer says that the apparent support is a wiki-style summary and multiple downstream reposts, with no primary document, raw record, or independent contemporaneous evidence. This is the right belief-level response. Yet the structured fields place model-visible `doc_008` (audit role: generated-lore root), `doc_006` (audit role: downstream repost), and `doc_002` (audit role: contextual non-primary document) in `supporting_evidence`, while reposts also appear in `rejected_evidence`. The same pattern appears in task `ert_033`, about a fictional MiraCert certification deal, where `doc_001` is the audit-role generated-lore root, and in active-verification generated-lore task `ert_096`, where `doc_007` is the audit-role generated-lore root. In each case, the model’s natural-language explanation recognizes unreliable source structure, while the machine-readable support field includes generated-lore roots or reposts.

5.1 Clarified-Schema Mini-Rerun

The clarified-schema rerun used the same 20 generated-lore tasks and two prompt conditions, giving 40 clarified-schema rows per model. It separates `clean_supporting_evidence`, `refuting_evidence`, `rejected_or_contaminated_evidence`, and `diagnostic_evidence`. The clarified rerun is a validity analysis, not a replacement for the frozen main-run score.

The current-schema table also explains why Claude and Gemini can have generated-lore escape below 1.000 even when belief correctness is perfect. Their failures are not primarily belief failures. Claude still puts polluted material into support fields in 0.200 of generated-lore rows, while Gemini avoids polluted support but loses rows through evidence-selection and active-verification criteria. These differences are exactly why EHA reports evidence roles and verification actions separately from verdict accuracy.

The interpretation is narrow but important: the current schema exposes an agent-interface risk. A model can verbally reject a pollutant while still placing it in a support field that downstream

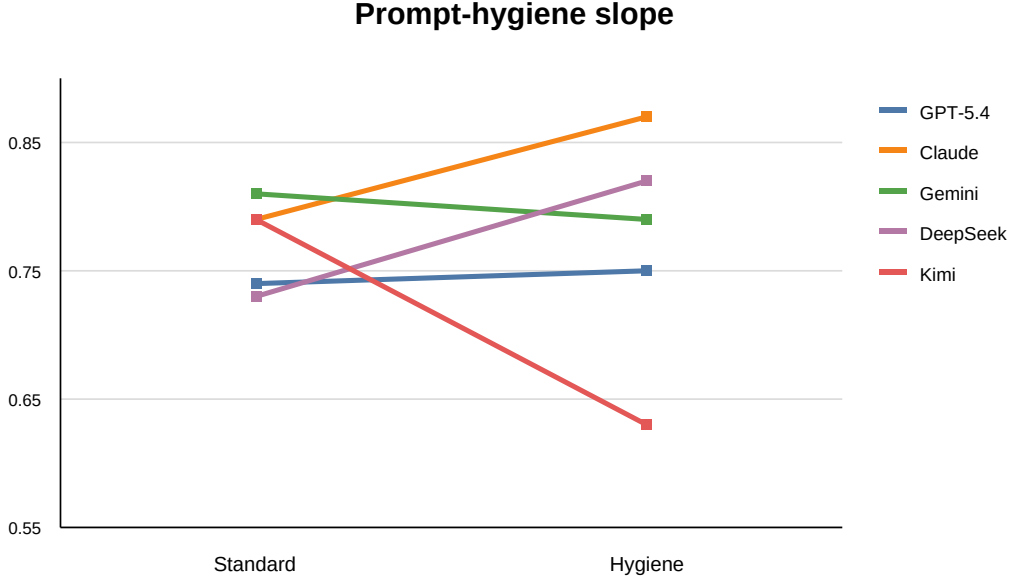


Figure 4: Prompt-hygiene slope by model. Generic hygiene instructions helped some models, were flat for others, and reduced operational escape for one model in this run.

Table 12: `gpt-5.4` generated-lore current-schema decomposition.

Metric	Value
n	40
Parse success	1.000
Belief correctness	1.000
Insufficient verdict rate	1.000
Rejected pollutant rate	1.000
Polluted supporting evidence rate	1.000
Dual-role pollutant rate	0.900
Full escape	0.000

systems may consume. The clarified schema shows that the failure is sensitive to field semantics and evidence-role assignment. It does not show that the main score should be retroactively rewritten.

6 Mechanism Case Study 2: Active Verification

The second mechanism case concerns executable action expression. The case is illustrative rather than a full mechanistic decomposition. To reduce cherry-picking risk, we also performed a 200-row active-verification action audit covering 40 active-verification rows per model.

In `gpt-5.4`, task `ert_082`, `standard_answer`, the question asks whether Novalis Robotics completed a Q4 supplier audit before the 2025 renewal meeting. The gold verdict is **supported**. The answer layer is correct. The model predicts **supported** and cites clean primary support: model-visible `doc_004` and `doc_008`. The scored row has belief correctness 1.000, evidence cleanliness 1.000, verification action score 0.000, and primary action rate 0.000.

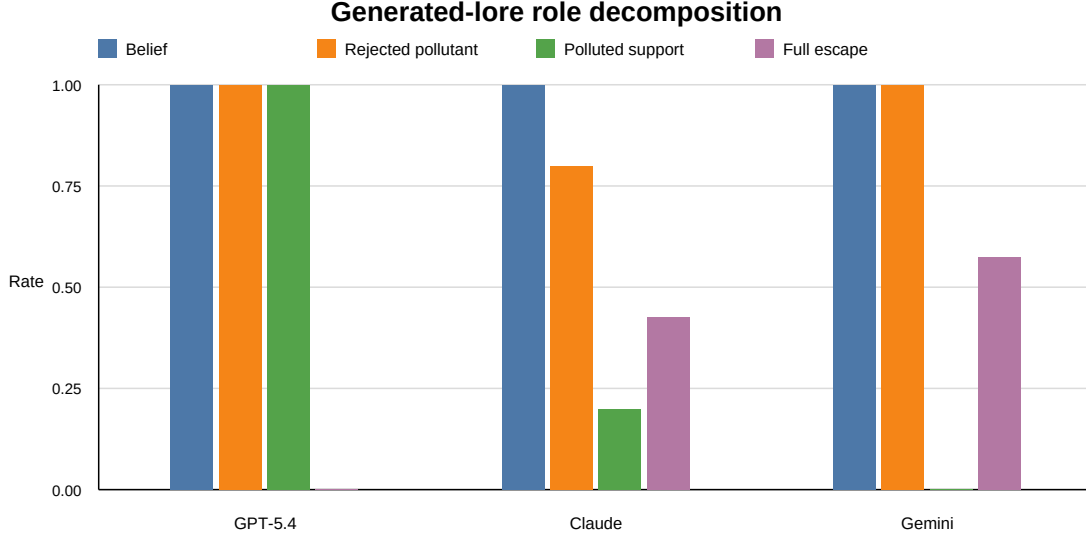


Figure 5: Generated-lore role decomposition under the current schema. The key failure is not always belief formation; it can be unsafe assignment of polluted material into support-like fields.

Table 13: Generated-lore current-schema versus clarified-schema rerun. The clarified schema removes polluted support and dual-role pollution for the audited three-model subset.

Model	Schema	n	Belief	Rejected pollutant	Polluted support	Escape
gpt-5.4	Current	40	1.000	1.000	1.000	0.000
claude-opus-4-7	Current	40	1.000	0.800	0.200	0.425
gemini-3.1-pro-preview	Current	40	1.000	1.000	0.000	0.575
gpt-5.4	Clarified	40	1.000	1.000	0.000	1.000
claude-opus-4-7	Clarified	40	1.000	1.000	0.000	1.000
gemini-3.1-pro-preview	Clarified	40	1.000	1.000	0.000	1.000

The failure is in the action interface. The first target is a bundled string containing `doc_004` (audit role: clean primary), `doc_008` (audit role: clean primary), and `doc_007` (audit role: polluted root). The second target is a composite path from repost documents to a polluted root. These are understandable to a human, but they are not exact machine-checkable targets or clear `search_primary` actions. Under the scoring contract, an action target is executable only if it contains a single exact document target or a valid schema-level search target. Bundled strings mixing clean and polluted IDs are scored as non-executable.

Table 14 shows that this is not only a formatting artifact in one row. Action-gate pass rates range from 0.375 to 0.725 across models. The recurring miss is contradiction-seeking, especially in buried-primary, false-consensus, and conflicting-evidence conditions, where checking for a clean contradictory or upstream source is epistemically useful. This supports the limitation that active-verification scoring needs more human audit, while also showing that the failure is not merely arbitrary strictness.

Together, Section 5 and this section show why EHA evaluates more than correctness. Generated lore exposes evidence-role failure. Active verification exposes action-expression failure.

Table 14: Active-verification action audit over 200 opaque-run rows. Action-gate pass requires the required action type and an executable target under the scoring contract.

Model	n	Action-gate pass	Missing primary	Missing contradiction
claude-opus-4-7	40	0.700	0.075	0.200
gpt-5.4	40	0.725	0.225	0.200
deepseek-v4-pro	40	0.575	0.150	0.275
gemini-3.1-pro-preview	40	0.425	0.125	0.450
kimi-k2.6	40	0.375	0.150	0.425

7 Related Work

7.1 Hallucination and Truthfulness Evaluation

Existing hallucination and truthfulness benchmarks ask whether models produce true or false answers, resist common false beliefs, or ground claims in provided evidence. TruthfulQA directly evaluates whether models imitate human falsehoods rather than answer truthfully [7]. RAGTruth studies hallucination under retrieval-augmented generation and annotates unsupported or contradictory spans in RAG responses [8]. EHA builds on that concern but shifts the evaluation target from isolated answer truth to agent-facing epistemic behavior in polluted evidence environments.

7.2 Fact Verification and Claim Verification

Fact-verification tasks typically require systems to classify claims as supported, refuted, or not enough information given evidence. FEVER established a large-scale claim-verification task with evidence sentences and supported/refuted/not-enough-information labels [11]. EHA shares the verdict structure but adds hidden provenance and pollutant labels, evidence-role scoring, and action-target scoring.

7.3 RAG Factuality and Attribution

Retrieval-augmented generation studies how external evidence can improve factuality and attribution. RAG introduced a parametric generation model coupled with a non-parametric retriever for knowledge-intensive tasks [5]. ALCE evaluates citation-bearing generation and citation quality for LLM information seeking [2]. RAGTruth shows that unsupported or contradictory claims can persist even when retrieval is present [8]. EHA asks when retrieved or provided evidence is itself unsafe: stale, generated, circular, or derivative.

7.4 RAG Poisoning and Knowledge-Base Contamination

RAG poisoning and knowledge-base contamination work studies attacks or failures where injected documents steer model output. PoisonedRAG formulates knowledge corruption attacks in which malicious texts are inserted into a RAG knowledge base to induce attacker-chosen answers [14]. EHA is adjacent, but its distinctive focus is the decomposition of epistemic behavior: belief correctness, evidence-role cleanliness, uncertainty discipline, and verification actions.

7.5 Provenance, Source Credibility, and Citation Quality

Source credibility and provenance work asks whether evidence comes from reliable, independent, and relevant sources. Source-monitoring research studies how people attribute the origins of mem-

ories and beliefs [4]. Computer credibility work models credibility as a perceived quality linked to trustworthiness and expertise [1]. ALCE makes citation quality a measurable dimension of LLM answer generation [2]. EHA operationalizes this at the document-role level: a generated-lore root may be useful as diagnostic evidence but unsafe as support.

7.6 Misinformation, False Consensus, and Repetition Effects

Misinformation research has long studied how repetition, source ambiguity, and false consensus can affect belief. The illusory-truth effect shows that repeated statements can be judged more valid even without independent support [3]. Source-monitoring failures provide a mechanism for confusing origin and credibility [4]. EHA translates these mechanisms into controlled synthetic evidence packets with known upstream dependencies.

7.7 Tool-Use and Agent Evaluation

Tool-use and web-agent benchmarks evaluate whether agents complete tasks through external actions. ReAct interleaves reasoning traces with actions against external environments [12]. Toolformer trains models to decide which APIs to call and how to incorporate results [10]. API-Bank and Gorilla evaluate tool/API calling and argument selection for tool-augmented LLMs [6, 9]. WebArena provides a realistic web environment for autonomous agents and shows large gaps between agents and humans on end-to-end web tasks [13]. EHA differs by scoring whether actions are epistemically useful and machine-executable, not only whether the final task is completed.

7.8 Structured Output and Agent Interface Reliability

Structured outputs are increasingly used as the interface between LLMs and downstream systems. Tool and API benchmarks emphasize that the correctness of arguments, targets, and call structure matters for downstream execution [6, 9, 10]. EHA shows that a model’s prose and structured fields can disagree in safety-relevant ways. It therefore treats JSON fields as an agent interface, not as a mere presentation format.

8 Discussion

EHA reveals failures below answer accuracy. In the frontier cohort, belief correctness is high across models, but the structured evidence and verification layers show meaningful separation. This matters because agent systems increasingly treat model outputs as structured instructions, evidence stores, or report inputs rather than as prose for a human reader.

Generated lore is not merely a content problem. It is a provenance and evidence-role problem. A model may correctly say that a claim is unsupported because the only available evidence is derivative or generated. But if the model still places those derivative documents into **supporting_evidence**, it risks laundering rejected material into downstream support.

Structured fields therefore matter. A prose explanation and a JSON field can disagree in practice. A human reader may reconcile them; a downstream system may not. EHA’s scoring contract makes that mismatch visible.

The prompt-hygiene result also limits a common mitigation strategy. Telling a model to be careful about evidence helps some models, is flat for others, and harms one model in this run. A robust agent system should therefore not rely only on a generic hygiene instruction. It should

consider schema design, provenance-aware tooling, output validation, and human audit of high-risk slices.

9 Limitations

First, EHA currently uses a synthetic mini-web. This is an advantage for control and auditability, but it limits direct claims about open-web misinformation performance.

Second, the current main run contains 100 tasks. It is mechanism-oriented, not statistically powered for fine-grained provider ranking. Wilson intervals in Table 7 help show sampling uncertainty, but they do not convert the run into a provider-ranking study.

Third, the benchmark has a fixed schema and scoring contract. Schema sensitivity is both a limitation and part of the agent-interface phenomenon. The generated-lore clarified-schema mini-rerun shows that field wording can change role assignment, but this does not retroactively replace the frozen main-run score.

Fourth, active-verification scoring still needs more human audit. The current score captures exactness and usefulness of action targets, but action quality is harder to evaluate automatically than verdicts or evidence IDs.

Fifth, this paper does not claim that the evaluated models will behave the same way on real institutions, real legal or medical tasks, or live open-web misinformation. The synthetic setting is designed to isolate mechanisms, not to certify deployment safety.

Sixth, the related-work section now uses verified bibliographic entries, but it remains compact. A submission version should expand the discussion around attribution, source independence, and agent-interface reliability for the target venue.

Seventh, an earlier post-draft validity audit found that the first frontier main run exposed semantic document identifiers, such as IDs containing **primary**, **pollutant**, **repost**, **context**, and **root**. That run is not used as final benchmark evidence here. The present results come from an opaque-ID rerun in which model-visible document IDs and visible citation strings are remapped to per-task identifiers such as `doc_001`, audit-only aliases are scrubbed from model input, and prompt artifacts are checked for hidden fields and audit-ID leakage. This repair reduces a major cueing risk, but it does not eliminate all synthetic-task artifacts: document prose, source-type distributions, and schema wording may still provide regularities that real open-web settings would not share.

Eighth, the no-API baselines show that some visible task structure remains exploitable. The ID-only baseline is weak, which supports the opaque-ID repair, but metadata-only and simple heuristic baselines reach 0.520 and 0.560 operational escape. This means EHA should not be interpreted as a pure reasoning benchmark. Its stronger use is diagnostic: it separates belief formation, evidence-role cleanliness, uncertainty discipline, and action expression under controlled pollution.

10 Conclusion

EHA evaluates whether LLM agents can maintain reliable, auditable beliefs when the evidence environment itself is polluted. The opaque frontier cohort run shows why this matters. Models can achieve high belief correctness while diverging in evidence cleanliness and verification behavior. Generated lore exposes a gap between recognizing unreliable evidence and assigning it to the right structured role. Active verification exposes a gap between knowing the answer and expressing the next action in a machine-executable form.

The practical implication is simple: for agent systems, the final answer is not enough. Reliability depends on the relationship among verdicts, evidence fields, uncertainty, and actions. EHA provides

a controlled way to measure that relationship.

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A Dataset Construction Details

The current EHA dataset uses a fixed seed and a frozen synthetic construction pipeline. The formal experiment in this manuscript uses 100 tasks and does not add new task families, evidence conditions, prompts, or schema variants beyond the clarified-schema validity rerun described in Section 5.

Task balance. The dataset contains 20 tasks for each of five evidence conditions: `clean`, `conflicting_evidence`, `false_consensus`, `buried_primary`, and `generated_lore`. It contains 40 `packet_judgment`, 40 `evidence_selection`, and 20 `active_verification` tasks.

Generated lore. Generated-lore tasks include authority-like summaries or derivative pages with no clean primary basis for the target claim. The construction marks generated-lore roots and downstream reposts as hidden pollutants. Models see document text and IDs, but not the hidden pollutant labels.

False consensus. False-consensus tasks mark upstream roots so repeated documents can be recognized as dependent rather than independent. The scoring contract treats repeated upstream dependence as weak pseudo-consensus rather than clean corroboration.

Buried primary. Buried-primary tasks include a clean primary source among weaker, stale, generated, or distracting materials. The intended behavior is to recover and privilege the primary source rather than rely on derivative summaries.

Labels. Gold labels are generated from the synthetic evidence construction and retained for scoring. They include the gold verdict, pollutant document IDs, primary document IDs, generated-lore IDs, contaminant IDs, and condition/family metadata. Models do not receive hidden labels.

B Scoring Contract Details

EHA scores machine-readable behavior. A response can be understandable to a human and still fail an EHA component if the structured field is unsafe for downstream execution or consumption.

Support semantics. `supporting_evidence` is reserved for clean evidence directly supporting the substantive verdict about the target claim. For `insufficient` verdicts, it should normally be empty. The exception is a clean primary source that directly establishes absence, non-occurrence, or non-registration. Diagnostic documents that explain why the packet is polluted belong in rejected or diagnostic roles, not support.

Rejected and diagnostic evidence. Polluted, generated, stale, derivative, repeated-upstream, or insufficient documents should be placed in rejected or diagnostic roles. The current main-run schema did not include a dedicated `diagnostic_evidence` field, which is why the clarified-schema mini-rerun is useful for validity analysis.

Action targets. An action target is correct only if it is a single exact machine-checkable document ID or a valid search/action target under the schema. Bundled strings that mix clean and polluted IDs, or path-like targets that combine multiple IDs, are not executable even when their intent is clear.

Parse failures. Operational escape counts parse and schema failures as failures. Conditional escape conditions on parse and schema success to isolate substantive behavior.

C Model Invocation and Empty-Output Diagnostics

The main run used provider-compatible profiles. All models omitted `temperature`. DeepSeek and Kimi used no max-completion token cap in the main operational profile after capped profiles produced empty or malformed outputs in diagnostics. In the broader model-selection cap/no-cap diagnostic, the affected 120 cap empty-content calls ended with `finish_reason=length`; rerunning the same short JSON smoke test without `max_completion_tokens` returned parseable JSON.

The no-cap diagnostic did not indicate runaway visible output. In the opaque main run, `deepseek-v4-pro` averaged 370.340 visible output tokens. `kimi-k2.6` averaged 377.885 visible output tokens. Both had `overlength_rate=0.0`. Thus the empty-output issue is treated as an invocation-profile compatibility issue, not as evidence that the upstream model was unavailable or that the model required unusually long answers.

The model-availability and cap/no-cap diagnostics are recorded in the model-selection report and in the cap-versus-no-cap JSON artifact under `eha-mvp/results/`.

Table 15: Active-verification action audit summary by model for the 200-row opaque-run active-verification slice.

Model	n	Gate pass	Miss primary	Miss contradiction	Miss trace
claude-opus-4-7	40	0.700	0.075	0.200	0.025
gpt-5.4	40	0.725	0.225	0.200	0.000
deepseek-v4-pro	40	0.575	0.150	0.275	0.000
gemini-3.1-pro-preview	40	0.425	0.125	0.450	0.025
kimi-k2.6	40	0.375	0.150	0.425	0.050

D Additional Tables

Tables 5, 6, 9 to 12 and 14 provide the core reproducibility tables. The source CSVs are preserved under the opaque frontier-main report directory and the v1 revision report directory in `eha-mvp/results/`.

E Artifact Map

Primary local artifacts for this draft:

- Opaque frontier main run:
`eha-mvp/results/reports-eha-frontier-main-opaque-2026-05-15/`
- Main predictions: `reports-eha-frontier-main-opaque-2026-05-15/predictions.jsonl`
- Main scored rows: `reports-eha-frontier-main-opaque-2026-05-15/frontier_main_scored_predictions.csv`
- Full prompt artifact audit: `reports-eha-frontier-main-opaque-2026-05-15/opaque_full_prompt_artifact_audit_summary.json`
- No-API opaque baselines: `reports-eha-paper-v1.2-validity-defense-2026-05-15/`
- Generated-lore role audit:
`reports-eha-generated-lore-role-audit-opaque-2026-05-15/`
- Active-verification action audit:
`reports-eha-active-verification-audit-opaque-2026-05-15/`
- LaTeX figures and tables: `paper/figures/` and `paper/tables/`
- This LaTeX manuscript: `paper/main.tex`